

Plugging into the Future: An Exploration of Electricity Consumption Patterns

TEAM ID	SWTID-2026-7684
PROJECT TITLE	Plugging into the Future: An Exploration of Electricity Consumption Patterns
MAX MARKS	3 MARKS

3.4 – Technology Stack

The project **"Plugging into the Future: An Exploration of Electricity Consumption Patterns"** was developed using a combination of hardware and software technologies to ensure efficient data processing, analysis, visualization, and web deployment.

Hardware Requirements

Processor: Intel Core i5

RAM: 8 GB or above

Storage: 256 GB SSD or higher

System Type: 64-bit Operating System

Internet Connection: Required for dashboard access and deployment

Software Requirements

Operating System: Windows 10/11

Programming Language: Python

Database: MySQL

Data Visualization Tool: Tableau

Web Framework: Flask

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Frontend Technologies: HTML, CSS, Bootstrap

Version Control: GitHub

Development Environment: Jupiter Notebook / VS Code

Web Browser: Google Chrome, Microsoft Edge, or Mozilla Firefox

Purpose of Technologies

Python was used for data cleaning, preprocessing, and analysis.

MySQL was used to store and manage electricity consumption data.

Tableau was used to create interactive dashboards and visualizations.

Flask was used to integrate the dashboard into a web application.

HTML, CSS, and Bootstrap were used to design and develop responsive web pages.

GitHub was used for source code management and version control.

This technology stack provided a reliable and scalable environment for analysing electricity consumption patterns and delivering interactive insights through a web-based dashboard.

BRAVO
— BUILDERS —
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